Design and Realization of Mixed-Technology Systems

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1. Unsyllabus

Name	Description
Course	M 1.9 Concept Engineering Mixed-Technology Systems (CEMS)
	M 1.10 Design and Realization of Mixed-Technology Systems (DRMS)
Term	Winter
Instructor	Prof. Dr.-Ing. M. Meiners, Dipl.-Ing. (FH) T. Ziemann

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The course makes heavy use of circuit simulation, using **Xschem** for schematic entry and **ngspice** for simulation. The 130nm CMOS technology **SG13G2** from IHP Microelectronics is used.

Tools and PDK are integrated in the **IIC-OSIC-TOOLS** Docker image, which will be used during the coursework.

All course material is made publicly available and shared under the Apache-2.0 license.

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2. Introduction and Survey

2.1 Course Objectives

- Interfacing Microsystems
  - Mixed-technology systems (System-on-Chip, SoC)
  - System analysis
  - System specification
- Design Methodology
  - Seamlessly modeling and design over all physical domains
- PCB (System) and IC Design
  - Architecture
  - Partitioning
  - Layout

2.2 Scientific Computing

- Python (Anaconda)
- Matlab (Campus Lizenz)
- Command-line tools

2.3 EDA Tools

- PCB / System Design
  - LTspice
  - KiCad EDA
  - Altium Designer
  - SiemensEDA PCB tools
  - cadence System Design & Analysis
- IC / Silicon Design
  - IIC-OSIC-TOOLS (open-source)
  - SiemensEDA IC tools
  - cadence IC Design & Verification
  - synopsys silicon design (IC)

2.4 OS-Tools

- Microsoft-Terminal
- Microsoft-PowerShell
- MacOS-Terminal
- Linux/MacOS Shell zsh-tools,
- git (Versionskontrolle)

2.5 Code Editors

- Visual Studio Code
- Spyder IDE
- Thonny (Micro-)Python IDE

- Emacs
- Vim

2.6 Data Science

- Journaled File System
  - Directories
  - Files
- Rectangular Data
  - Comma-Separated-Values (CSV), Tab-Separated-Values (TSV)
  - csvkit: A suite of utilities for converting to and working with CSV, the king of tabular file formats.
  - [miller: Miller is like awk, sed, cut, join, and sort for name-indexed data such as CSV, TSV, and tabular JSON] (https://github.com/wireservice/csvkit)
  - Spreadsheet (.xlsx, .ods)

2.7 Publish Computational Content

- quarto
  - Manuscripts
- Jupyter-Book

2.8 Are you writing or TeXing?

- MikTeX (Windows, MacOS, Linux)
- MacTeX (MacOS)
- TeXLive (Linux)

2.9 LaTeX Editors

- IDE's
  - TeXStudio
  - TeXMaker
- Collaborative Frameworks
  - Overleaf, Online LaTeX
  - CoCalc - Online LaTeX

2.10 Bibliography and LaTeX

- Citavi
- Zotero
- JabRef

2.11 Model-Based Systems Engineering (MBSE) of an Inertial Sensor System and IC Design

This winter term, the defining idea of the CEMS course is a modular board-level sensor system consisting of ADXL335 accelerometer, ADS1115 ADC and ESP8266 NodeMCU. The ADC module has to be replaced by a custom IC design.

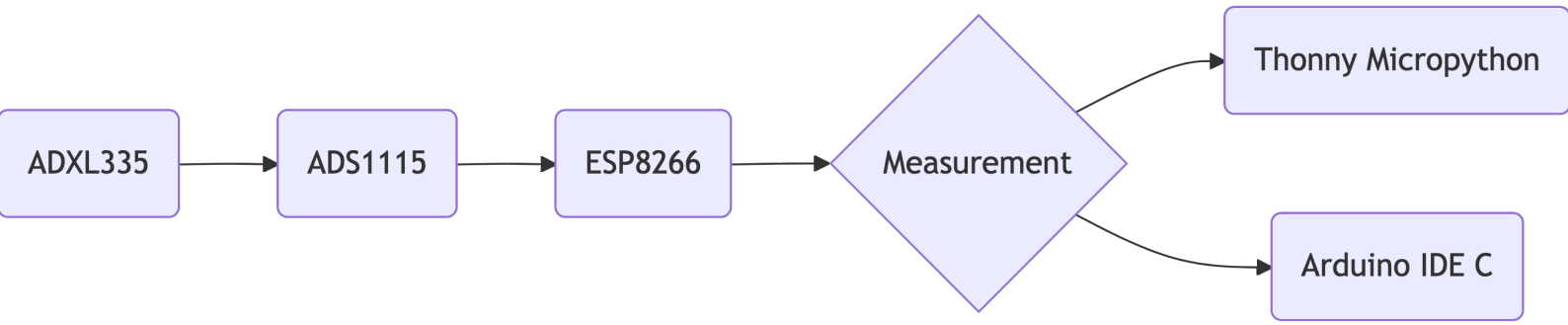


Figure 2.1: Modular board-level sensor system.

- System level, behavioural model
  - Matlab/Simulink,
  - Python
- PCB level
  - ESP8266 NodeMCU,
  - TIs ADS1115,
  - ADs ADXL335
- IC level, SPICE with behavioural blocks, e.g. OTA and comparator
  - IIC-OSIC-TOOLS IHP130-based analog and digital chip design
- **Final Oral Exam/Project Presentation**

2.12 Course Prerequisites

- Fundamentals of linux operating systems
- Fundamentals of microelectronics
  - Device physics and models
  - Transistor level analog circuits, elementary gain stages
- Fundamentals of analog circuit design
  - Operational amplifier
  - Active filter design
  - Noise analysis
  - Switched-capacitor techniques
- Prior exposure to SPICE, Matlab, Python or equivalent.

Bibliography

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